

Single-phase off-grid inverter

GPEO-4KLI/6KL1/12KL1

High-efficiency power generation

- The maximum conversion efficiency is 99.9%
- A wide MPPT voltage range of 85V to 450V
- Single-channel high current charging up to 27A
- Up to 12 units can be paralleled for flexible expansion

Safe and reliable

- IP54 protection grade, reliable quality
- Super overload capacity (200%/5S)
- 3-year warranty, more peace of mind when using

User-friendly

- Touch keys, full-view LCD display
- Intelligent noise reduction mode, enjoy the quiet moments
- The body is simple, thin and light, and easy to install
- The battery has strong compatibility and can be flexibly configured for storage

Intelligent operation and maintenance

- Intelligent backup load and generator interfaces make electricity usage more worry-free
- Intelligent terminal management enables real-time control of system data
- Supports OTA remote upgrade and Settings, with low operation and maintenance costs

UBETTER

Product model	GPEO-4KL1	GPEO-6KL1	GPEO-12KL1
Dc input (photovoltaic)			
Maximum recommended string power	4000W	6000W	12000W
Maximum input voltage	500V	500V	500V
MPPT operating voltage range	85V-450V (@85V starting voltage)	85V-450V (@85V starting voltage)	85V-450V (@85V starting voltage)
Quantity of MPPT	1	1	2
The number of MPPT strings in a single channel	1	2	2
Maximum current of a single MPPT channel	15A	27A	27/27A
Single-channel MPPT short-circuit current	18A	35A	35/35A
Ac input (power grid)			
Maximum output power	5000W	11500W	16000W
Maximum output current	28A	50A	90A
Rated grid voltage	220/230/240Vac	220/230/240Vac	220/230/240Vac
Grid frequency	50/60Hz	50/60Hz	50/60Hz
Grid voltage range	170-280Vac(UPS mode) 90-280Vac(Load mode)	170-280Vac(UPS mode) 90-280Vac(Load mode)	170-280Vac(UPS mode) 90-280Vac(Load mode)
Battery parameters (Bidirectional)			
Battery type	Lithium iron phosphate/lead-acid	Lithium iron phosphate/lead-acid	Lithium iron phosphate/lead-acid
Battery voltage range	20~30Vdc	40~60Vdc	40~60Vdc
Rated battery voltage	24Vdc	48Vdc	48Vdc
Maximum charge/discharge current	120A/160A	120A/130A	210A/230A
Battery BMS communication	RS485	RS485	RS485
Ac output (off-grid)			
Rated output power/apparent power	4000W/4000VA	6000W/6000VA	12000W/12000VA
Rated output current	18.2A	27.3A	54.5A
Rated output voltage	220/230/240Vac 50/60Hz	220/230/240Vac 50/60Hz	220/230/240Vac 50/60Hz
Parallel machine capability	Cannot be paralleled	It supports up to 12 units in parallel at most	It supports up to 6 units (single-phase) in parallel and 12 units (three-phase) in parallel
Instantaneous peak output power	8000VA 5s	12000VA 5s	24000VA 5s
Total harmonic distortion of current (@Linear load)	<3%	<3%	<3%
Switching delay		10ms(UPS); 20ms(Load)	
Efficiency			
Peak efficiency	95%	98%	98%
Maximum power of MPPT	99.90%	99.90%	99.90%
Protection			
Built-in protection/detection function	√Output short-circuit protection √Output overvoltage protection		
Surge protection	Photovoltaic end: Type III, AC end: Type III		
IP protection level	IP54		
Conventional parameters			
Operating temperature range	-10℃~55℃		
Relative humidity range	5%~95%	5%~95%	5%~95%
Job height/Demotion	A deduction of over 2000m	A deduction of over 2000m	A deduction of over 2000m
Standby loss	<10W	<10W	<20W
Installation method	Wall-mounted	Wall-mounted	Wall-mounted
Heat dissipation method	Air-cooled	Air-cooled	Air-cooled
Communication	RS232/RS485/Dry contact/Wi-Fi	RS232/RS485/Dry contact/Wi-Fi	RS232/RS485/Dry contact/Wi-Fi
Display	LCD	LCD	LCD
Size and weight			
Inverter size (L*W*H)	347*445*120	347*445*120	525*630*125
Net weight of the inverter (kg)	9.2	12.4	28.5
Packaging size (L*W*H)	560*465*240	560*465*240	745*650*240
Gross weight (kg)	11.4	14.6	32
Warranty period	Three years		